

Error analysis of generating 3D models of objects using various hardware systems.

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Abstract—The article is devoted to a comparative analysis of software and hardware systems for creating small-sized three-dimensional (3D) models. The paper discusses various methods for generating and evaluating the quality of 3D objects, such as photogrammetry, TrueDepth(FaceID), and LiDAR. It also offers a classification of 3D objects by size, which will help in choosing the appropriate methods of analysis and model generation. The article is intended for experts, students, visualization specialists and those interested in 3D content.

Keywords— *three-dimensional models; computer graphics; photogrammetry; TrueDepth; LiDAR; metrology 3D objects.*

I. INTRODUCTION

Intensively developing technologies for the effect of immersing the viewer in media content (virtual reality technologies, interactive films, films using computer graphics, etc.), digitalization of medicine and other fields of activity stimulate the development of a large number of methods for generating 3D content. Research and analysis of new and existing methods, and their comparison with each other, is important for choosing the optimal method for generating 3D models. When choosing a method, you need to take into account the scale, texture, grid of the model, and other parameters of the objects being modeled. In this article, we will describe the difference between 3D models created using different 3D model generation methods: photogrammetry, made from different cameras, and compare them with those created by laser scanning (LiDAR), FaceID scanning (TrueDepth).

Research objectives:

1. To study the theoretical foundations of photogrammetry and the principle of operation of photogrammetric methods.
2. Consider existing technologies for creating 3D models.
3. Conduct an experiment on creating 3D models using various technologies (cameras, LiDAR, TrueDepth, iPhone 15 Pro max).
4. Analyze the results obtained and evaluate the accuracy and detail of the 3D models.
5. Identify the advantages and disadvantages of each technology.
6. To formulate the main conclusions of the study and determine the prospects for the development of photogrammetric technologies.

II. THE THEORETICAL PART

To talk about the accuracy of 3D model generation methods, a general classification of objects by size should be introduced. At the moment, there are many classifications from different fields with duplicate names and overlapping ranges.

Some of them are:

1. Geometry and Engineering

Nanoscale: less than 100 nm (nanometers).

Micro-sizes: from 100 nm to 1 mm (micrometer).

Miniature sizes: from 1 mm to 10 cm (centimeters).

Average size: from 10 cm to 1 m (meter).

Large size: over 1 m.

2. Construction and architecture

Small-size structures: up to 1 m.

Medium-sized structures: from 1 m to 10 m.

Large-size structures: over 10 m.

3. Natural objects

Micro-objects: plants and animals up to 1 cm in size.

Mesoobjects: plants, animals, rocks, etc. size from 1 cm to 1 m.

Macro objects: plants, animals, landscape, reservoirs, etc. size over 1 m.

Most modern methods of generating 3D objects allow you to create models from a few millimeters to several hundred meters. The classification does not include ranges other than this.

The main classification range is "*Objects*" – objects ranging from 30 cm to 3 m – these are the main objects of the human environment and interaction: human-human, human-tool, human-animal, human-vehicle, human-room, etc. Therefore, they are most in demand when building a virtual simulation.

We have placed three more objects up and down from this range: *milli-*, *santi-*, *deci-*, *kilo-*, *mega-*, and *gigaobjects*, on a logarithmic scale relative to the *Objects*.

1. *Milliobjects* less than 0.003 m (nuts, coins, accessories, etc.[4])

2. *Centiobjects* from 0.003 m to 0.03 m (tools, jewelry, etc.)

3. *Deciobjects* from 0.03 m to 0.3 m (stationery, large jewelry, cutlery, tableware, accessories)

4. *Objects* from 0.3 m to 3m (decorative elements, gadgets, clothes, shoes, sculpture, architectural elements, etc.)

5. *Kiloobjects* from 3 m to 30 m (transport, architectural and painting objects, large interior details)

6. *Megaobjects* from 30 m to 300 m (buildings and premises [6])

7. *Gigaobjects* over 300 m (landscapes, drilling rigs, quarries, excavations)

An overview of existing technologies for creating 3D models.

Below is a brief description of a number of widely used 3D model generation technologies with their advantages and limitations. We will pay special attention to photogrammetry, which we have chosen as the optimal technology available at the moment, working with objects of different sizes, textures and properties.

1. Laser scanning (3D scanning).

A method based on using a laser rangefinder to measure distances to an object and create an accurate three-dimensional model of it. The laser beam is reflected from the surface of an object, fixing its position in space. The scanners create a high-resolution point cloud, which can then be transformed into a 3D model[7].

Advantages:

- High precision and detail for both *Milliobjects* and *Gigaobjects*;

- Low light requirements;

- High-speed processing of scan results.

Disadvantages:

- High cost of equipment;

- Long scan time;

- The need for special software for data processing and highly qualified operators.

2. LiDAR (Light Detection and Ranging).

The laser-based technology will be used to scan objects and determine their shape, size, and position in space. It is widely used in vehicle autopiloting. The position of the object is determined by measuring the time interval between the probing and reflected pulse.

Advantages:

- High accuracy of scanning from *Objects* to *Mega* objects;

- Low cost of equipment (relative to the cost of laser scanning);

- Low light requirements.

Disadvantages:

- High degree of dependence on the accuracy of equipment characteristics;

- Low accuracy when working with complex surfaces;

- High sensitivity to weather changes.

3. FaceID (TrueDepth),

a technology used in Apple devices to identify users by facial parameters, which allows you to create 3D models of objects using a camera and other sensors developed for FaceID:

TrueDepth is a camera system consisting of: conventional and infrared cameras, as well as a spot projector. A point projector projects an infrared grid of 173x173 points onto an object, and an infrared camera captures the curvature of the point and thus determines the shape of the object. A regular camera generates textures for the 3D model in question.

Advantages:

- Ease of use;

- Fast model generation (no more than 10 seconds).

Disadvantages:

- Inhomogeneous density of points on the final model due to the parallelism of the projector's grid rays;

- Low detail accuracy;
- Available only on Apple devices.

4. RGB-D cameras.

Using RGB-D cameras (for example, Microsoft Kinect) allows you to get both color data and depth information in real time. They are widely used in the gaming industry. This makes them useful for quickly creating simple 3D models.

Advantages:

- Availability of devices and software;
- Fast model generation.

Disadvantages:

- Low accuracy and detail.

5. CT scan. Computed tomography, which creates a series of slices of an object, which are then combined into a 3D model. It is used for scanning human or animal organs, as well as any objects whose integrity should not be compromised.

Advantages:

- The ability to visualize the internal structures of objects;
- Fast scanning process.

Disadvantages:

- Radiation exposure (may be dangerous if used regularly);
- High cost of equipment;
- Complexity of maintenance and operation;
- Inability to work with metal objects.

6. Contact scanning.

Contact scanning methods use mechanical devices to measure distances to the surface of an object. For example, coordinate measuring machines (CMMs) can be used.

Advantages:

- High measurement accuracy.

Disadvantages:

- Low scanning speed;
- The risk of damage to the object during interaction.

7. Manual modeling

In this method, artists or engineers create 3D models from scratch using computer graphics software (for example, Blender, Autodesk Maya, 3ds Max). They can be based on photographs or drawings of real objects.

Advantages:

- Full control over the model creation process.

- The ability to create complex shapes and details.

Disadvantages:

- Long terms of preparation and production of models.
- High requirements for the training and qualifications of specialists.

8. Photogrammetry.

A method for creating three-dimensional models of objects based on the analysis of a large number (more than 40) two-dimensional images. For this purpose, photographs of the object taken from different angles are used. Special software, for example, Meshroom, Agisoft Metashape, Autodesk ReCap Pro and others, process images, find common points and create a point cloud, which is then used to build a 3D model[1].

Advantages:

- Accessibility: regular photo and video cameras can be used;
- High level of detail (at high resolution from *Millionobjects* to *Mega* objects).

Disadvantages:

- A large number (from 40, depending on the required accuracy) of images are required to obtain a high-quality model.;
- Poor quality in low light conditions or in the presence of reflective or transparent surfaces.

Analysis of photogrammetry workflow parameters.

Photogrammetric methods are based on the idea of comparing two or more images of the same object taken from different points of view. This allows you to get information about the shape, size, and position of an object in space[8, 10].

1. Camera calibration

Calibration is an important step, which includes setting camera parameters (focal length, distortion parameters, etc.). This is necessary to ensure the accuracy of measurements and to prevent distortion in the resulting images.

- Focal length of the lens

The optimal focal length is 35-50 mm (full frame equivalent) to minimize distortion. It is undesirable to use wide-angle lenses, as they can cause distortion at the edges of the frame.

- Resolution

The minimum is 12 megapixels (MP), but above 20 MP is recommended.

Higher resolution allows you to capture small details of an object, which increases the accuracy of the three-dimensional model.

For example, for *Objects* (up to 3 meters), 20 MP and higher are recommended.

For objects from *Kiloobjects*, it is preferable to use cameras of 36 MP and higher.

- ISO

It is advisable to keep the ISO as low as possible (within 100-400) to reduce noise in photos. The noise should not exceed one degree of ISO gradation.

2. Data collection

The photogrammetry process begins with capturing images of a target object or area. Photo cameras, drones, satellites, or other devices capable of taking photos from various angles can be used for this purpose[9].

- Number of photos.

Simple objects: 40-70 photos for objects of simple shape.

Standard objects: 70-300 photos for objects of medium complexity.

Complex objects: from 300 photos and more.

A large number of photos increases the chances of successful matching of key points and minimizes errors.

- Frame overlap.

The overlap should be 60-80% of the frame area to ensure coherence and accuracy of the model. Each part of the object should be visible in at least 3-5 photos. Insufficient overlap can lead to broken links between images, loss of model accuracy, and errors.

- Geometric distribution.

The intervals between the camera angles should be equal, both horizontally and vertically (for example, every 10-15 degrees), to create a model with a uniform grid and without artifacts[7].

- Illumination.

Lighting should be uniform and standard in all photos to avoid shadows and overexposure on the final textures. The brightness should be at least 80 cd/m² and no more than 150 cd/m² or 500 lux, which corresponds to standard office lighting. Unstable lighting can cause distortions and errors in estimating the depth of the model.

- Color temperature.

It is necessary to maintain a constant color temperature of the light source.

Daylight has a color temperature of about 5500 K. When using artificial lighting, it is necessary to adhere to the temperature range of 4500-6500 K.

- Minimum distance to the object

Close-up shooting: 0.1-1 meter distance from a million objects and less to objects.

Standard shooting: Kiloobjects with a distance of 1-5 meters.

Remote shooting: from Mega objects and more, the distance is 5-20 meters.

Close-up shooting can cause excessive perspectives and distortions that degrade the quality of the model.

- Control points (markers)

Place markers evenly throughout or around the object.

The distance between markers for *Deciobjects* and less and About 5 – 20 cm from each other, for *Objects* and *Kilobjects* and 50-100 cm for *Megaobjects* and more is at least 5-10% of the object size[9].

Markers help you accurately determine the scale and orientation of an object in space.

3. Image processing

At this stage, special software algorithms are used to detect common points (Agisoft Metashape, VisualSFM) in different images. The software, based on computer vision techniques, analyzes images to identify feature points that can be compared between different frames[9].

The program performs the following tasks:

- Alignment of photos relative to each other.
- Removal of distortions caused by the curvature of the Earth and other factors.
- Creating a dense point cloud, which is a set of coordinates of points on the surface of an object.

4. Creating a three-dimensional model

Using trigonometric calculations and photogrammetry methods, the spatial coordinates of matching points are calculated (the Delaunay or least squares triangulation method). Based on the data obtained, a point cloud is formed, which is a set of three-dimensional coordinates. The object model can be represented as a polygonal grid, a set of surfaces, or a solid body.

5. Post-processing

The process of texturing a 3D model takes place through the correlation of object boundaries and the gradient of colors in photographs.

6. Verification and analysis

After receiving the model, its accuracy is checked by comparing it with reference data or other sources of information. The quality of the model is assessed based on various criteria, including accuracy, resolution, and smoothness.

III. THE EXPERIMENTAL PART

As part of our experiment, 3D models were created using the following methods: photogrammetry, LiDAR, and FaceID(TrueDepth).

The Meshroom v.2023 software was used for photogrammetry.3.0 and pre-installed in the Apple iPhone 15 Pro max. The latter was also used for LiDAR and TrueDepth.

The images for photogrammetry were taken with cameras with the characteristics described in Table 1.

Model	Resolution	Focal Length	Aperture	Pixel resolution in photo	Type of matrix
A31-48M-R03 QK11 08 (Samsung)	48Mp	25mm	F/2.0	4000x2250	6.4 x 4.8 mm
Sony IMX803	48Mp	24mm	F/1.78	5712x4284	CMOS (6.4 x 4.8 mm)
Sony IMX714(TruDepth)	12.19Mp		F/1.9	4032x3024	CMOS BSI
Sony ILCE-6000	20Mp	27mm	F/4.0	6000x3376	APS-C (23.5 x 15.6 mm)

Table 1 — Camera characteristics

Photogrammetry is a method of constructing 3D models based on photographs taken from various positions around an object [5]

Evaluation of the accuracy and detail of 3D models.

Using the example of two objects, a peach and a clock, we will demonstrate the quality assessment process used in this study. In the end, we will summarize the evaluation data of twenty models (Table 3).

Consider the methods and approaches used to evaluate the quality of models created by the six 3D model generation methods mentioned earlier.

The choice of software and hardware complexes (SHC) was determined by their prevalence and accessibility. The sample does not include Chinese SHCs due to their abundance, which requires a separate study dedicated to them.

Meshroom, Sony (M-So) — the model was created using photogrammetry on the Meshroom software, and the photos on the Sony ILCE-6000 camera.

Meshroom, Samsung (M-Sa) — the model was created using photogrammetry using Meshroom software, and the photos were taken with a Samsung A31-48M-R03 camera QK1108 .

Meshroom, Apple (M-A) — the model was created using photogrammetry using Meshroom software, and the photos were taken with a Sony IMX803 camera.

Apple, Apple (A-A) — The model was created using photogrammetry on Apple software, and the photos were taken on a Sony IMX803 camera.

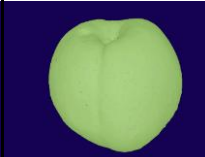
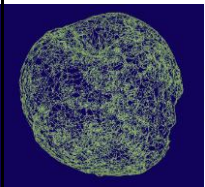
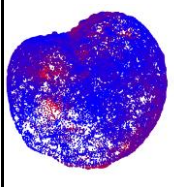
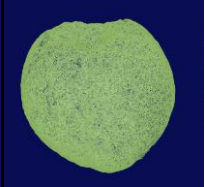
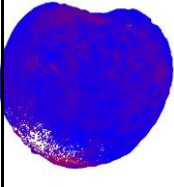
TrueDepth (TD) — the model was created using FaceID technology using Apple software, and the photos were taken with a Sony IMX714 camera.

LiDAR (LD) — the model was created using LiDAR from Apple.

Table 2 shows illustrations of the final models (Viewport), mesh density and uniformity (Mesh), and the Mesh comparison method (MCM) score.

The Mesh comparison method evaluates the deviation of the spatial values of the studied model from the reference one, focusing on the coordinates of each specific point of the 3D model grid. A positive deviation is highlighted in red, and a negative deviation is highlighted in blue.

To apply this method, a reference model is needed with which all subsequent models will be compared. As a reference model in each specific series, we selected the model with the highest subjective rating, which is shown below (Table 3).

	Viewport	Mesh	MCM
Original object			
M-So			reference model
M-Sa			
M-A			

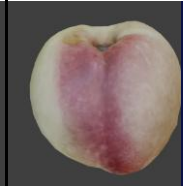
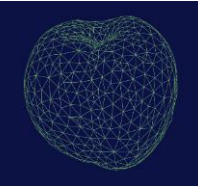
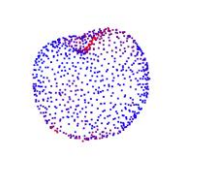
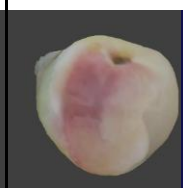
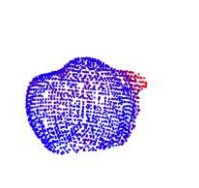
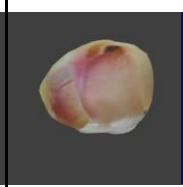
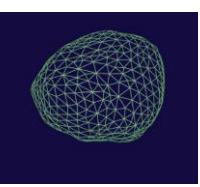
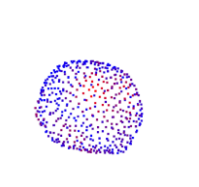
A-A			
TD			
LD			

Table 2 — Illustrations of the studied models and the original object

Subjective assessment of quality

For the subjective evaluation of models, five criteria are identified: uniformity of the grid, the presence of artifacts, the size and number of grid breaks, the quality of the UV scan (texture distribution according to the 3D model), and similarity to the original object.

Each of these criteria was evaluated from 1 to 5 points.

Uniformity of the grid

1-the grid points are not connected and are unevenly spaced from each other, 5-all grid points are connected to neighboring ones and spaced equidistant from each other.

The presence of artifacts

1-the entire surface of the model is noisy, there are more than 10 artifacts, 5-the surface of the model is not noisy, artifacts are not present.

The size and number of grid gaps

1-gaps make up more than a third of the visible part of the model, and their number is more than 3, 5-there are no gaps in the grid

UV scan quality (texture distribution according to the 3D model)

1-the texture is blurred/distorted, 5- complete similarity to the original object.

Similarity to the original object

1-The 3D model does not resemble the original, the object is not recognized, 5- complete similarity to the original object (the model image can be mistaken for a real object)

The respondents for this assessment were twelve experts with professional skills in 3D graphics and visual content.

Mesh comparison method

MCM- illustrates the deviation of two three-dimensional models relative to each other. The coordinates of each point of the models in the Cartesian coordinate system (X, Y, Z) are used.

MCM is used by archaeologists and reconstructionists to achieve an optimal ratio of model weight to detail [2, 3, 4].

For all its simplicity, MCM has a number of limitations. It only works with vertices and does not take edges and faces into account, which leads to missing artifacts that occur on them. MCM also requires the use of an “initial” quality model that will serve as a reference, which is not always possible.

Below are the MCM graphs presented as a model observation system (in billions) from tracking the probability of its distribution (Figures 1-4).

To improve the readability of the graphs, the values from 0 mm to 0.2 mm are combined.

As a mathematical model for the nectarine, the M-tak is taken in conjunction with the best indicator of the subactive function, for the M-A clock (Table 3).

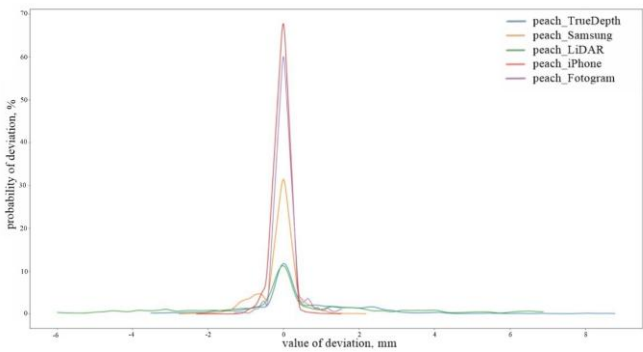


Fig. 1 — approximating graphical layout for the peach (nectarine) model

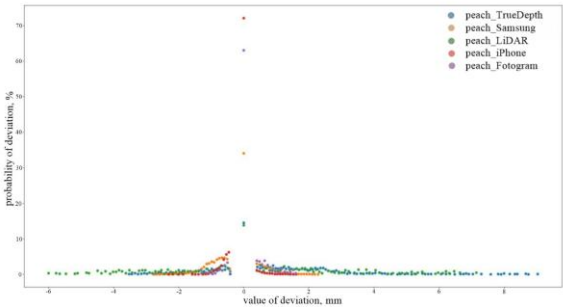


Fig. 2 — original graphical layout for the peach (nectarine) model

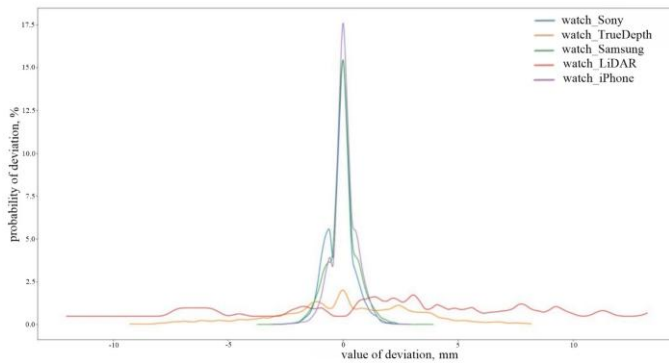


Fig. 3 — Approximating graphical layout for clock models

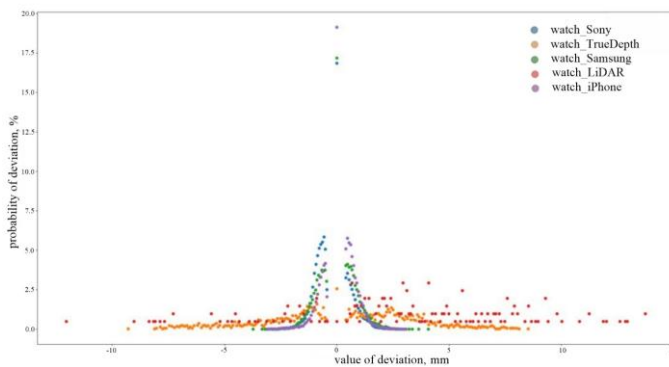


Fig. 4 — Original graphic layout for watch models

Based on the graphs, you can roughly estimate the quality - the accuracy of the match of the compared models with the reference one. As you can see, the researchers are colonic and are co-authors of the normal distribution (raspredeleni Gausa), which guides the use of methods of probability theory and statistics to analyze the quality of 3D models. Just as in the case of video distribution, it is of great importance to you, first of all, 0 mm and something more than bank indicators (krill), the more it concerns your partner. The more uniform the characteristic, the lower the noise level.

Arithmetic mean

The arithmetic mean deviation (AMD) indicator, which we defined as the arithmetic mean (1) deviation from the total value of the estimated model in microns.

$$x = 1/n \sum_{i=1}^n x_i = 1/n(x_1 + \dots + x_n) \quad (1)$$

somewhere there is a connection and two points of view on the fact that I am in a billion, but something in common with the orienting model.

Standard deviation of the sample

The sample standard deviation (SDS) parameter is a partial standard deviation indicator, which is usually determined by the sum of the indicators characterizing the i-th value and the AMD value that mattered for all:

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - x_{AAD})^2}{(n-1)}}, \quad (2)$$

somewhere one is a connection and there are two points of view on the fact that I am in a billion, but something in common with the orienting model.

Average distortion relative to the original

The parameter of the average distortion relative to the original (ADRO) as the arithmetic mean (1) of the maximum lengths of 3D models to the maximum lengths of the original (real) objects as a percentage along the X, Y, Z axes (3).

$$xi(x,y,z) = (L_{3D}(x,y,z)/L_{RO}(x,y,z) - 1)100\%, \quad (3)$$

here L_{3D} is the maximum length of a 3D object along one of the X, Y, Z axes, L_{RO} is the maximum length of a relational object along one of the X, Y, Z axes.

Method of creating model	AMD, mm	SDD, mm	ADRO, %	Subjective assessment
nectarine (peach)				
M-So	0	0	1.995	4.6
M-Sa	-0.245	0.723	2.567	3.5
M-A	-0.202	0.327	2.234	3.9
A-A	0.140	0.503	1.475	4.5
TD	0.843	1.934	7.657	3.2
LD	0.344	2.758	9.795	1.8
watch				
M-So	-0.213	0.798	4.672	4.2
M-Sa	0.033	0.888	6.679	3.2
M-A	0.161	0.788	6.359	3.4
A-A	0	0	5.199	4.5
TD	0.296	3.123	13.933	2.8
LD	3.139	4.837	29.732	1.8
average characteristics for all 20 models				
M-So	-0.092	0.682	3.829	3.95
M-Sa	-0.227	0.754	6.512	3.17
M-A	0.103	0.713	6.875	3.32
A-A	-0.106	0.702	4.301	3.87
TD	2.023	2.286	11.042	2.14
LD	4.59	4.728	22.874	1.42

Table 3 — Generalized quality characteristics of all models.

IV. COMPARATIVE ANALYSIS

Based on the study of twenty 3D models, the resulting table was compiled (Table 4), which lists the methods of generating 3D models in order of changing their quality assessment from higher to lower (where 1 is the best method).

Table 4 shows that, apparently, all you need is a method, which is used in the Meshroom v.2023.3.0 program, autographs are produced from a Sony ILCE-6000 camera.

It's just that in practice, M-Luchier told you so when he was working with the scale of objects, and when Luchier did, he told you A-A. M-Sa and M-A suggested changes directly at the same level.

TD and LD have done very little work with millions of users, and for this, do not try to work with large amounts of information that you can find here, and they themselves will show you the best.

Quality assessment	AMD	SDD	ADRO	Subjective assessment
1	A-A	M-So	M-So	M-So
2	M-So	A-A	A-A	A-A
3	M-Sa	M-A	M-Sa	M-A
4	M-A	M-Sa	M-A	M-Sa
5	TD	TD	TD	TD
6	LD	LD	LD	LD

Table 4 — The resulting table

V. CONCLUSION

First of all, you can use this application to set up graphics and hand clocks, which allows you to get a complete report on the quality and speed of creating 3D models.

The variety of settings and the complexity of the characteristics of modern photo and video equipment does not allow us to unambiguously determine the best models for

photogrammetry. A lot depends on production needs, availability, and other factors. It seems to be the most effective use of benchmarks to select equipment for existing tasks.

We hope that our research will help to better define the parameters that are important for the selection of the method and equipment.

During the research, we tested the existing methods of generating 3D models, their advantages and disadvantages. Excellent experience in creating 3D models, measuring everything and quality.

I plan to look at the methods of genealogy of 3D models with a stand from Coloprojects and more, as well as the introduction of artificial intelligence into these models.

Further development of 3D model generation methods involves increasing their accuracy and accessibility, as interest in them is growing due to the development of immersive technologies, which in turn increases the amount of content for training neural networks and their subsequent application to improve 3D model generation methods.

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